

Topic 1.4

1.4: C++ Operators

The division operator / behaves as follows:

int / *int* produces an *int* that is the *whole* part of the quotient.

int / *float* produces a *float*

float / *int* produces a *float*

float / *float* produces a *float*

The ?: operator is *triadic* - it requires **3** operands:

```
(expr) ? (value if expr is true) : (value if expr is false)
```

For example, instead of writing

```
if (x > y)
    larger = x;
else
    larger = y;
```

we can write

```
larger = (x > y) ? x : y;
```

Casts are used to force the compiler to perform a type change. This can remove some safety checks, so use these carefully:

```
int i;
float x;

x = i;      /* OK because all ints can be represented as floats */
i = x;      /* Causes compilation error because not all
             floats are representable as ints. */

i = (int) x; /* OK because you are forcing compiler to convert.
             This may cause i to have an "interesting"
             value if x is not in the range representable as
             ints. */
```

to topic 1.5 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-1-dot-5>)